

Stat 207 HW4

1. X : # defectives in daily production of product

Y : a day's production

θ : probability that any product is defective

$$X|Y=y \sim \text{Binomial}(y, \theta) \rightarrow \binom{y}{x} \theta^x (1-\theta)^{y-x}$$

$$Y \sim \text{Poisson}(\lambda) \rightarrow \frac{\lambda^y e^{-\lambda}}{y!}$$

$$\theta|Y=y \sim \text{Beta}(\alpha, \gamma) \rightarrow \frac{\theta^{\alpha-1} (1-\theta)^{\gamma-1}}{B(\alpha, \gamma)}$$

$$\star \pi(\theta|y) = \frac{\pi(\theta) f(y|\theta)}{\int \pi(\theta) f(y|\theta) d\theta}$$

$\star$ we observe x , but not y
 $\rightarrow$ results should not have y

$$\star \binom{n}{k} = \frac{n!}{k!(n-k)!}$$

a) Show that $X|\theta \sim \text{Poisson}(\lambda\theta)$

$$f(x|\theta) = \sum_y f(x|y) f(y|\theta)$$

$$= \sum_{y=x}^{\infty} \underbrace{\binom{y}{x} \theta^x (1-\theta)^{y-x}}_{\text{Binomial}} \underbrace{\frac{\lambda^y e^{-\lambda}}{y!}}_{\text{Poisson}}$$

$$= e^{-\lambda} \theta^x \sum_{y=x}^{\infty} \frac{\binom{y}{x} (1-\theta)^{y-x} \lambda^y}{y!} = e^{-\lambda} \theta^x \sum_{y=x}^{\infty} \frac{(1-\theta)^{y-x} \lambda^y}{y!}$$

$$\frac{e^{-\lambda} \theta^x}{\lambda^{-x} (x!)} \sum_{y=x}^{\infty} \frac{((1-\theta)\lambda)^{y-x}}{(y-x)!} \rightarrow \frac{e^{-\lambda} \theta^x}{x!} e^{\lambda(1-\theta)} = \frac{e^{-\lambda+\lambda-\lambda\theta} (\lambda\theta)^x}{x!}$$

$$\rightarrow \boxed{P(X|\theta) = \frac{(\lambda\theta)^x e^{-\lambda\theta}}{x!}} \sim \text{Poi}(\lambda\theta)$$

b) Yes, θ is independent of y . The number of products produced is independent of the probability of a product being defective. α and γ are free of y .

$$c) f(\theta|x) \propto f(x|\theta) \cdot \pi(\theta) = \frac{\lambda^x \theta^x e^{-\lambda\theta}}{x!} \cdot \frac{\theta^{\alpha-1} (1-\theta)^{\gamma-1}}{B(\alpha, \gamma)} \propto \exp(-\lambda\theta) \theta^{x+\alpha-1} (1-\theta)^{\gamma-1} \quad \checkmark, 0 < \theta < 1$$

d) Obtain full conditionals $\underbrace{y|x=x, \theta}_{\text{if}}$ $\underbrace{\theta|x=x, y=y}_{\text{if (beta result)}}$

$$i. P(y=y|x=x, \theta) \propto P(x=x, y=y|\theta) = \underbrace{P(x=x|y=y, \theta)}_{\binom{y}{x} \theta^x (1-\theta)^{y-x}} \cdot \underbrace{P(y)}_{\frac{e^{-\lambda} \lambda^y}{y!}} = \frac{\theta^x (1-\theta)^{y-x} e^{-\lambda} \lambda^y}{x! (y-x)!} = \boxed{\frac{e^{-\lambda} \lambda^x (1-\theta)^{y-x} \lambda^{y-x}}{x! (y-x)!}}$$

$$ii. P(\theta|x=x, y=y) \propto \underbrace{[x=x|\theta, y=y]}_{\binom{y}{x} \theta^x (1-\theta)^{y-x}} \cdot \underbrace{[\theta|y]}_{\theta^{\alpha-1} (1-\theta)^{\gamma-1}}$$

$$= \binom{y}{x} \theta^x (1-\theta)^{y-x} \theta^{\alpha-1} (1-\theta)^{\gamma-1} \propto \theta^{x+\alpha-1} (1-\theta)^{y-x+\gamma-1}$$

$$\boxed{\text{Beta}(\alpha+x, \gamma+y-x)}$$

e) Assume independent prior on λ according to which $\pi(\lambda) \propto 1, \lambda > 0$

Full conditionals $\lambda | \theta, x$ and $\theta | \lambda, x$ are they proper?

Obtain joint posterior $\pi(\lambda, \theta | x)$, is this proper?

$$e) \lambda, \theta | x \\ = \frac{\lambda^x \theta^{x-1}}{\Gamma(x)}$$

$\propto$ some function

$g(\lambda, \theta)$, is it

proper? (integral $\int_0^\infty \int_0^1$)

$$\int_0^\infty \int_0^1 g(\lambda, \theta) d\theta d\lambda$$

$$i. f(\lambda | \theta, x) \propto f(\theta, \lambda | x) \cdot f(\lambda | x, \theta)$$

$$e^{-\lambda \theta} \theta^{x-1} (1-\theta)^{x-1} \cdot \lambda^x \theta^x e^{-\lambda \theta} = e^{-2\lambda \theta} \theta^{2x-1} (1-\theta)^{x-1} \lambda^x \\ = \theta^{2x-1} (1-\theta)^{x-1} \lambda^x e^{-2\lambda \theta}$$

Is it proper?

$$\propto \lambda^x e^{-2\lambda \theta}$$

$$\int_0^\infty \lambda^x e^{-2\lambda \theta} d\lambda$$

bounded between

1 and $e^{-2\lambda}$, $\lambda^x \rightarrow \infty$ faster than $e^{-2\lambda} \rightarrow 0$ as $\lambda \rightarrow \infty$

$\rightarrow \infty$ so the conditional is improper

$$ii. f(\theta | \lambda, x) \propto f(\theta, \lambda | x) \cdot f(\lambda | \theta, x)$$

$$\dots \text{same} \rightarrow \theta^{2x-1} (1-\theta)^{x-1} \lambda^x e^{-2\lambda \theta}$$

$$\lambda^x \theta^{2x-1} (1-\theta)^{x-1} e^{-2\lambda \theta}$$

Is it proper?

$$\lambda^x \int_0^1 \theta^{2x-1} (1-\theta)^{x-1} e^{-2\lambda \theta} d\theta \text{ this is not proper}$$

$$iii. f(\lambda, \theta | x) = e^{-\lambda \theta} \theta^{x-1} (1-\theta)^{x-1} \frac{(\lambda \theta)^x}{x!}$$

$$\int_0^\infty \int_0^1 \frac{(\lambda \theta)^x}{x!} \theta^{x-1} (1-\theta)^{x-1} d\theta d\lambda$$

$$f(\lambda, \theta | x) = \frac{f(\lambda, \theta, x)}{f(x)} \quad \pi(\lambda) \pi(\theta)$$

Let $y = (y_1, \dots, y_n)^T$ random sample from Cauchy $(0, 1)$, $\theta > 0$

2. Prove posterior $p(\theta|y) = \frac{\prod_{i=1}^n [1 + (\theta - y_i)^2]^{-1} \mathbb{I}(\theta > 0)}{\int_0^\infty \prod_{i=1}^n [1 + (\theta - y_i)^2]^{-1} d\theta}$ is proper

→ integrates to 1 (not infinite)

$\mathbb{I}(\theta > 0)$ strictly positive, gamma?

= 1 if $\theta > 0$, 0 otherwise

need $\int p(\theta|y) d\theta < \infty$

$$\int_0^\infty p(\theta|y) d\theta = \int_0^\infty \frac{\prod_{i=1}^n [1 + (\theta - y_i)^2]^{-1} \mathbb{I}(\theta > 0)}{\int_0^\infty \prod_{i=1}^n [1 + (\theta - y_i)^2]^{-1} d\theta} d\theta = \frac{\int_0^\infty \prod_{i=1}^n [1 + (\theta - y_i)^2]^{-1} d\theta}{\int_0^\infty \prod_{i=1}^n [1 + (\theta - y_i)^2]^{-1} d\theta}$$

$L(\theta)$

improper priors: $U(-\infty, \infty)$ Beta $(0, 0)$

Integral w/o normalizing constant $< \infty$

For proper priors, $\int_0^\infty \pi(\theta) d\theta = 1$ e.g. gamma, exponential, inverse-gamma

then we have

$$\int_0^\infty \prod_{i=1}^n [1 + (\theta - y_i)^2]^{-1} d\theta = \int_0^\infty \prod_{i=1}^n [1 + (\theta - y_i)^2]^{-1} d\theta$$

but for non-informative / flat / improper priors

2.

$$\int_0^\infty \prod_{i=1}^n \frac{1}{1 + (\theta - y_i)^2} \cdot 1 d\theta < \infty?$$

* always want to check this for

these types of priors, posterior

analysis needs to be valid

Sum this → look at just one y_i * factors are all fractions, so their product is too

$$\leq \int_0^\infty \frac{1}{1 + (\theta - y_i)^2} d\theta \quad * \int_0^\infty \frac{dt}{1+t^2} = \arctan t \Big|_0^\infty = \pi/2$$

$$t = \theta - y_i \\ dt = d\theta$$

$\lim_{t \rightarrow \infty} \arctan(t) = \frac{\pi}{2}$ which is finite, product of n finite values is also finite,

$$-\frac{\pi}{2} < \arctan(t) < \frac{\pi}{2}$$

∴ posterior is proper because integral is finite

$y_i \sim N(0, 1)$

$y_i \sim \text{Cauchy}(0, 1)$

3. Show $E_n[w(\theta)^2] = \infty$

$$w(\theta) = \frac{f(y|\theta)\pi(\theta)}{h(\theta)} \rightarrow [w(\theta)]^2 = \frac{[f(y|\theta)]^2 [\pi(\theta)]^2}{[h(\theta)]^2}$$

$$h(\theta) \sim N(4.45, 0.38)$$

$$\int_0^\infty [w(\theta)]^2 h(\theta) d\theta$$

show $\rightarrow \infty$ as $h \rightarrow \infty$ under normal $h(\theta)$

for some $M > 0$, $\theta_0 > 0$ s.t. $\forall \theta > \theta_0$, $[w(\theta)]^2 h(\theta) > M$

$$\int_{\theta_0}^\infty [w(\theta)]^2 h(\theta) d\theta \rightarrow \int_{\theta_0}^\infty \frac{f(y|\theta)^2 \pi(\theta)^2}{h(\theta)} d\theta \rightarrow \int_{\theta_0}^\infty \frac{f(y|\theta)^2}{h(\theta)} d\theta$$

$$M \int_{\theta_0}^\infty d\theta = \infty$$

$$\frac{1}{\sigma\sqrt{2\pi}} \int_{\theta_0}^\infty \frac{\exp\left\{-\frac{1}{2\sigma^2}(\theta - \mu)^2\right\}}{\prod_{i=1}^n [1 + (\theta - y_i)^2]} d\theta$$

$$\int_{\theta_0}^\infty \frac{\prod_{i=1}^n [1 + (\theta - y_i)^2]^{-2}}{\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2\sigma^2}(\theta - \mu)^2}} d\theta$$

numerator $\rightarrow \infty$ faster than denominator, so whole integral $\rightarrow \infty$



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4. Prove that in rejection sampling, samples are generated from $h(\theta) = \frac{L(\theta)\pi(\theta)}{\int L(\theta)\pi(\theta)d\theta}$

algorithm: generate $\theta_j \sim g(\theta)$

generate $U \sim \text{Uniform}(0,1)$

If $M U_g(\theta_j) \leq L(\theta_j)\pi(\theta_j)$, accept θ_j , otherwise reject θ_j

$M > 0$, smooth density $g(\theta)$ envelope function

θ_j * jth accepted observation

$$P(\theta_j \in A) = \int_A h(\theta) d\theta$$

from $h(\theta)$

$$P(\theta_j \in A | M U_g(\theta_j) \leq L(\theta_j)\pi(\theta_j))$$

prove

$$\frac{P(\theta_j \in A, M U_g(\theta_j) \leq L(\theta_j)\pi(\theta_j))}{P(M U_g(\theta_j) \leq L(\theta_j)\pi(\theta_j))}$$

def. of conditional probability

random variables

$$\begin{aligned} \rightarrow b &= \int P(U \leq \frac{L(\theta)\pi(\theta)}{M g(\theta)} | \theta) \cdot g(\theta) d\theta \\ &= \int \frac{L(\theta)\pi(\theta)}{M g(\theta)} \cdot g(\theta) d\theta \\ &= \frac{1}{M} \int L(\theta)\pi(\theta) d\theta \end{aligned}$$

since θ is a r.v.

uniform, integrating just gives RWS

will cancel w/ numerator a

$$\begin{aligned} \rightarrow a &= P(\theta_j \in A, M U_g(\theta_j) \leq L(\theta_j)\pi(\theta_j)) \\ &= \int_A P(U \leq \frac{L(\theta)\pi(\theta)}{M g(\theta)} | \theta) \cdot g(\theta) d\theta \\ &= \frac{1}{M} \int_A L(\theta)\pi(\theta) d\theta \end{aligned}$$

bring back together:

$$\frac{\frac{1}{M} \int_A L(\theta)\pi(\theta) d\theta}{\frac{1}{M} \int L(\theta)\pi(\theta) d\theta} = \int_A h(\theta) d\theta$$

Fundamental theorem of calculus, differentiate w.r.t θ

$$\rightarrow \frac{L(\theta)\pi(\theta)}{\int L(\theta)\pi(\theta) d\theta} = h(\theta) \checkmark$$